

# DATA CLEANING, ANALYSIS AND DATA VISUALIZATION USING EXCEL



Tools Used: Excel

# PROJECT OVERVIEW

This project aims to clean raw data manually, perform descriptive analysis on the data, and create a dashboard to understand the data visually.

## OBJECTIVES

- Cleaning data manually using Excel.
- Determining the item type with the highest sales.
- Knowing the profit trend over the years.
- Creation of KPIs to display the average retail sales.
- Creation of slicers and sparklines.
- Visualizing the operations made with pivot tables.
- Creation of a dashboard.

## DATASET

This is a secondary dataset from data.gov. It contains 307646 records with 9 features. The nine features include: Year, month, supplier, item code, item type, retail sales, retail transfers, and warehouse sales.

The target variable in this project is retail sales.

## TOOLS

- Excel

## DATA CLEANING

- Removed duplicates from the item code
- Removed columns with blanks
- Fixed data types
- Rearranged the columns
- Changed the item description column from text to columns
- Hid some of the columns that would not have much impact on the data.

# EXPLORATORY DATA ANALYSIS

By using pivot tables, the following are the analysis done:

## 1. Analysis of the sales over the years.

The charts showed that 2019 recorded the highest sales compared to the rest.

## 2. Pie Chart of Item Types

The item types visualized shows wine is the most sold item type.

## 3. Average Monthly Warehouse and Retail Sales

This analysis aimed to see the distribution of sales over the years.

# DASHBOARD

Dashboard elements:

- KPI Cards
- Sparkline
- Charts
- Pivot Tables
- Conditional Formatting
- Slicers



# REFERENCES

## Data Source

[https://catalog.data.gov/dataset/warehouse-and-retail-sales?  
from\\_hint=eyJxIjojU2FsZXMiLCJzb3J0IjoicmVsZXZhbmNlIn0%3D](https://catalog.data.gov/dataset/warehouse-and-retail-sales?from_hint=eyJxIjojU2FsZXMiLCJzb3J0IjoicmVsZXZhbmNlIn0%3D)

## Github Repository

[https://github.com/Janicekipkiror/Warehouse\\_and\\_Retail\\_Sales\\_Analysis/tree/main](https://github.com/Janicekipkiror/Warehouse_and_Retail_Sales_Analysis/tree/main)